

Namhyuk Ahn

ASSISTANT PROFESSOR AT INHA UNIVERSITY

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Research Interests

My research goal is to develop powerful generative AI (Gen AI) solutions applicable across various domains, creating a strong Gen AI+X framework. Currently, I aim at creating specialized methodologies to adapt large-scale foundational Gen AI models in fields such as media, art, healthcare, manufacturing, and logistics, although my scope is not limited to these areas. I am also interested in exploring the societal implications of Gen AI and developing ways to mitigate potential negative impacts that may arise from the development of Gen AI.

Work Experience

Inha University

ASSISTANT PROFESSOR

Incheon, Korea

Sep. 2024 - Present

- School of Electrical and Electronic Engineering
- Department of Electrical and Computer Engineering (Graduate School)

NAVER WEBTOON

AI RESEARCHER

Pangyo, Korea

Aug. 2021 - Aug. 2024

- Built user-centric AI tools designed for cartoon creators/artists (e.g. Cartooner, DreamStyler)
- Researched user privacy-aware Gen AI (e.g. Impasto, which prevents copyright violation by Gen AI models)
- Developed portrait stylization system (e.g. WebtoonMe)

NAVER AI LAB

VISITING RESEARCHER

Bundang, Korea

Sep. 2019 - Oct. 2020

- Researched data augmentation for image super-resolution (e.g. CutBlur)
- Developed label-efficient conditional generative models
- Co-worked with Jaeeun Yoo, Youngjung Uh and Yunjey Choi

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INTERN

Bundang, Korea

June 2018 - Aug. 2018

- Developed image-to-image translation pipeline for talking head project

Education

Ajou University

PH.D. IN ARTIFICIAL INTELLIGENCE

Suwon, Korea

Mar. 2016 - Aug. 2021

- Advisor: Prof. Kyung-Ah Sohn
- Thesis: Toward an Efficient Deep Image Restoration Method

Ajou University

BACHELOR OF MEDIA IN DIGITAL MEDIA

Suwon, Korea

Mar. 2012 - Aug. 2016

Selected Publications

- Soyoung Yoon, **Namhyuk Ahn**, and In Kyu Park. T2LF: LLM-Guided Multimodal Diffusion for Text-to-Light Field Synthesis. **WACV 2026 (accepted)**
- Sungnyun Kim, Junsoo Lee, Kibeom Hong, Daesik Kim, and **Namhyuk Ahn**. DiffBlender: Scalable and Composable Multimodal Text-to-Image Diffusion Models. **Expert Systems with Applications (ESWA) 2026 (accepted)**
- **Namhyuk Ahn**, Wonhyuk Ahn, KiYoon Yoo, Daesik Kim, and Seung-Hun Nam. Imperceptible Protection Against Style Imitation from Diffusion Models. **IEEE Transactions on Multimedia (TMM) 2026 (accepted)**
- **Namhyuk Ahn**, KiYoon Yoo, Wonhyuk Ahn, Daesik Kim, and Seung-Hun Nam. Nearly Zero-Cost Protection Against Mimicry by Personalized Diffusion Models. **CVPR 2025**

- Jihye Back, Namhyuk Ahn, and Jangho Kim. Magnitude Attention-based Dynamic Pruning. **Expert Systems with Applications (ESWA)**, 2025
- Namhyuk Ahn, Junsoo Lee, Chunggi Lee, Kunhee Kim, Daesik Kim, Seung-Hun Nam, and Kibeom Hong. DreamStyler: Paint by Style Inversion with Text-to-Image Diffusion Models. **AAAI 2024**
- Namhyuk Ahn, Jaejun Yoo, and Kyung-Ah Sohn. Data Augmentation for Low-Level Vision: CutBlur and Mixture-of-Augmentation. **International Journal of Computer Vision (IJCV)** 2024
- Jeong-Hyeon Moon*, Namhyuk Ahn*, and Kyung-Ah Sohn. Decomposing Texture and Semantics for Out-of-distribution Detection. **Expert Systems with Applications (ESWA)**, 2024
- Namhyuk Ahn, Patrick Kwon, Jihye Back, Kibeom Hong, and Seungkwon Kim. Interactive Cartoonization with Controllable Perceptual Factors. **CVPR 2023**
- Namhyuk Ahn, Byungkon Kang, and Kyung-Ah Sohn. Efficient Deep Neural Network for Photo-Realistic Image Super-Resolution. **Pattern Recognition (PR)** 2022
- Jaejun Yoo*, Namhyuk Ahn*, and Kyung-Ah Sohn. Rethinking Data Augmentation for Image Super-Resolution: A Comprehensive Analysis and a New Strategy. **CVPR 2020**
- Namhyuk Ahn, Byungkon Kang, and Kyung-Ah Sohn. Fast, Accurate, and Lightweight Super-Resolution With Cascading Residual Network. **ECCV 2018**

Professional Service

Reviewer CVPR, ICCV, ECCV, NeurIPS, ICLR, ICML, AAAI, ACMMM, WACV
TPAMI, IJCV, TIP, TMM, TCSVT, SPIC, ESWA, Neurocomputing

Editor Mathematical Biosciences and Engineering (2022-2023; Guest)

Teaching

2025-2 Deep Learning, Python Programming, Multimedia (English)

2025-1 Multimedia, Vision-Language Models (Graduate), Multimodal AI (Graduate)

2024-2 Computer Graphics, Creative Design for Engineering

2017 Deep Learning and its Applications (at FastCampus)

Awards

2024.1 **NAVER N Innovation Award**, 3rd prize on R&D track

2023.1 **NAVER N Innovation Award**, 2nd prize on R&D track

2018.6 **Honorable Mention Award**, SR Challenge on NTIRE workshops @ CVPR

Invited Talks

2025.10 **ICCE-Asia (Tutorial)**, Responsible Generative AI: From Provenance to Protection

2025.9 **SNU Hospital**, Useful or Harmful? Rethinking Generative AI in Vision

2025.7 **KCC**, Content Protection in Generative AI Era

2023.3 **Ajou University**, Recent Advances in Visual Generative Models

Press Mentions

2024.4 **Digital daily**, “네이버웹툰, ‘작품 무단 학습 AI’ 막는 모델 선보인다”

2023.2 **Tech M**, “미국 · 유럽 · 일본도 ‘감탄’... 웹툰 AI 시대 열어야죠”

2023.1 **Digital daily**, “내 얼굴이 웹툰 캐릭터로 변했다”